CS3000 Practice Quiz 5D — Answer Key

Harder transfer: palindromes via LCS and reconciliation via edit distance

Notation follows the class notes on dynamic programming: $C(v)$ (coin change), $K(i, w)$ (0/1 knapsack), $L(i, j)$ (LCS), $D(i, j)$ (edit distance); strings and items are 1-indexed. Every table below was computed by a program and checked against brute force. Shaded cells mark the traceback path. Point splits imitate the Gradescope rubrics used on Quizzes 1–4 (each question 5 pts, 10 total).

Q1. Minimum insertions to make a palindrome

(a) Characters that already pair up correctly in the final palindrome form a subsequence that reads the same forwards and backwards, i.e. a common subsequence of s and s^R. Keep the longest such subsequence, $\ell = \mathrm{LCS}(s, s^R)$; every one of the other $n - \ell$ characters needs a partner inserted on the mirror side, and one insertion per unmatched character always suffices. So the minimum is $n - \ell$.

(b) Table (shaded = traceback path).

$s_i \downarrow$ $s_j^R \rightarrow$	0	1 (B)	2 (A)	3 (C)	4 (B)	5 (A)
0	0	**0**	0	0	0	0
1 (A)	0	0	**1**	**1**	1	1
2 (B)	0	1	1	1	**2**	2
3 (C)	0	1	1	2	**2**	2
4 (A)	0	1	2	2	2	**3**
5 (B)	0	1	2	2	3	**3**

(c) LCS = 3 (e.g. `ABA`), so $5 - 3 =$ **2** insertions. Example: `ABCAB`→`BACBCAB`. Check: `BACBCAB` reversed is `BACBCAB`. (Other answers such as `ABACABA` also work.)

(d) The LCS table is $(n + 1) \times (n + 1)$ with $O(1)$ work per cell: $\Theta(n^2)$ time and space.

Rubric (5 pts).

+1.5 pts Explains the matched-pairs / LCS argument (not just the formula).

+1.5 pts LCS table correct (-0.5 per wrong row).

+1 pt 2 insertions and a valid palindrome.

+1 pt $\Theta(n^2)$.

Q2. Bank reconciliation is edit distance

(a) $D(i, j)$ = fewest discrepancies explaining the difference between your first i transactions and the bank's first j. Same recurrence as edit distance:

$$D(i,j) = \begin{cases} j & i = 0 \\ i & j = 0 \\ D(i-1, j-1) & R[i] = B[j] \\ 1 + \min\big(D(i, j-1),\, D(i-1, j),\, D(i-1, j-1)\big) & R[i] \neq B[j] \end{cases}$$

Only in your record = deletion ($D(i-1, j)$); only at the bank = insertion ($D(i, j-1)$); different amounts = substitution ($D(i-1, j-1)$); equal amounts match for free.

(b) Table (shaded = traceback path).

$R_i \downarrow\ B_j \rightarrow$	0	\$18	\$7	\$48	\$20	\$9
0	**0**	1	2	3	4	5
1 (\$18)	1	**0**	1	2	3	4
2 (\$30)	2	**1**	1	2	3	4
3 (\$7)	3	2	**1**	2	3	4
4 (\$45)	4	3	2	**2**	3	4
5 (\$20)	5	4	3	3	**2**	**3**

(c) $D(5, 5) = \mathbf{3}$. From the traceback: \$18: same transaction; \$30: only in your record; \$7: same transaction; \$45 vs. \$48: different amount; \$20: same transaction; \$9: only on the bank statement.

Rubric (5 pts).

+1.5 pts Identifies edit distance and maps all three discrepancy types.

+2 pts Table correct (−0.5 per wrong row).

+1.5 pts 3 discrepancies, listed consistently with the table.